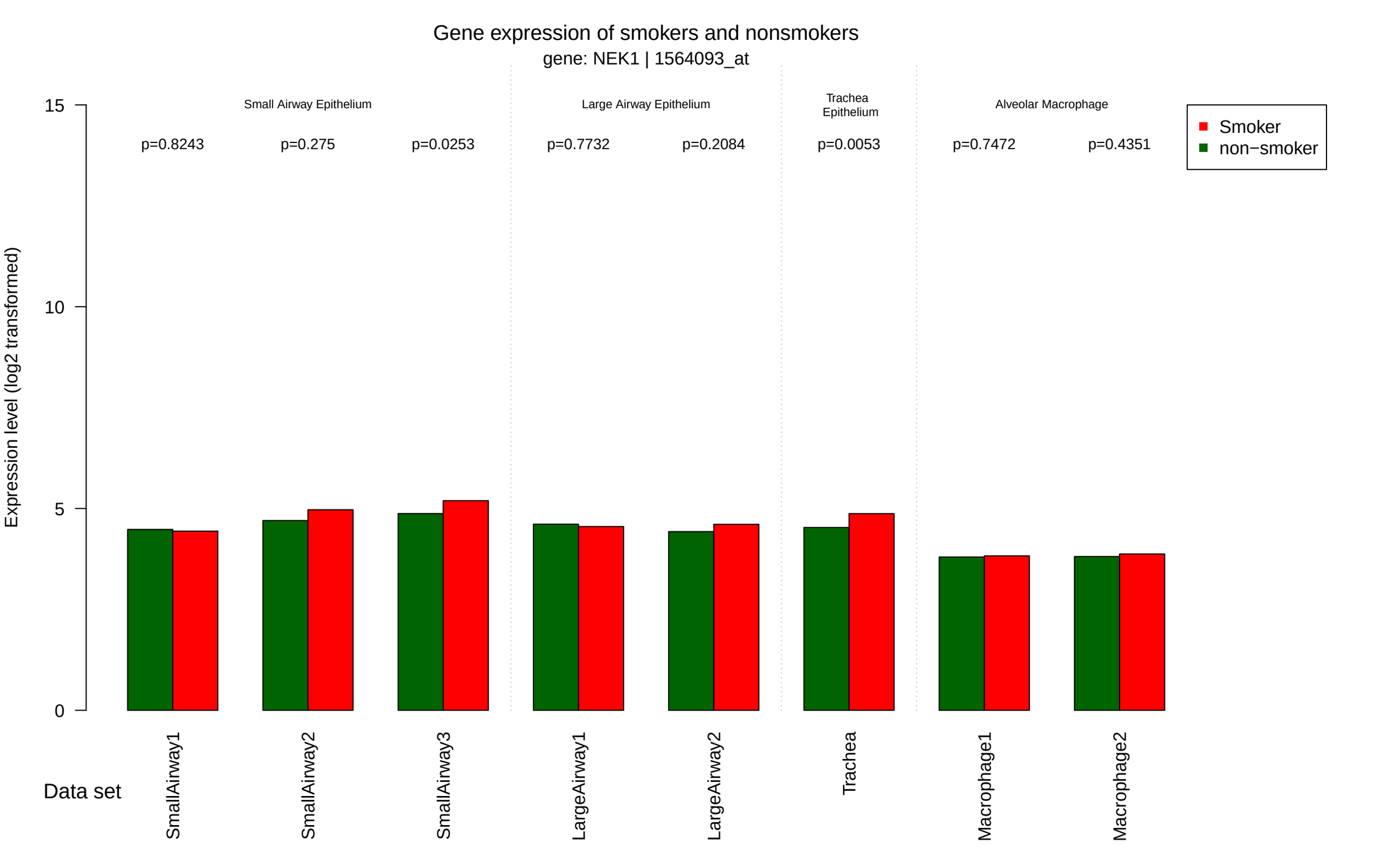
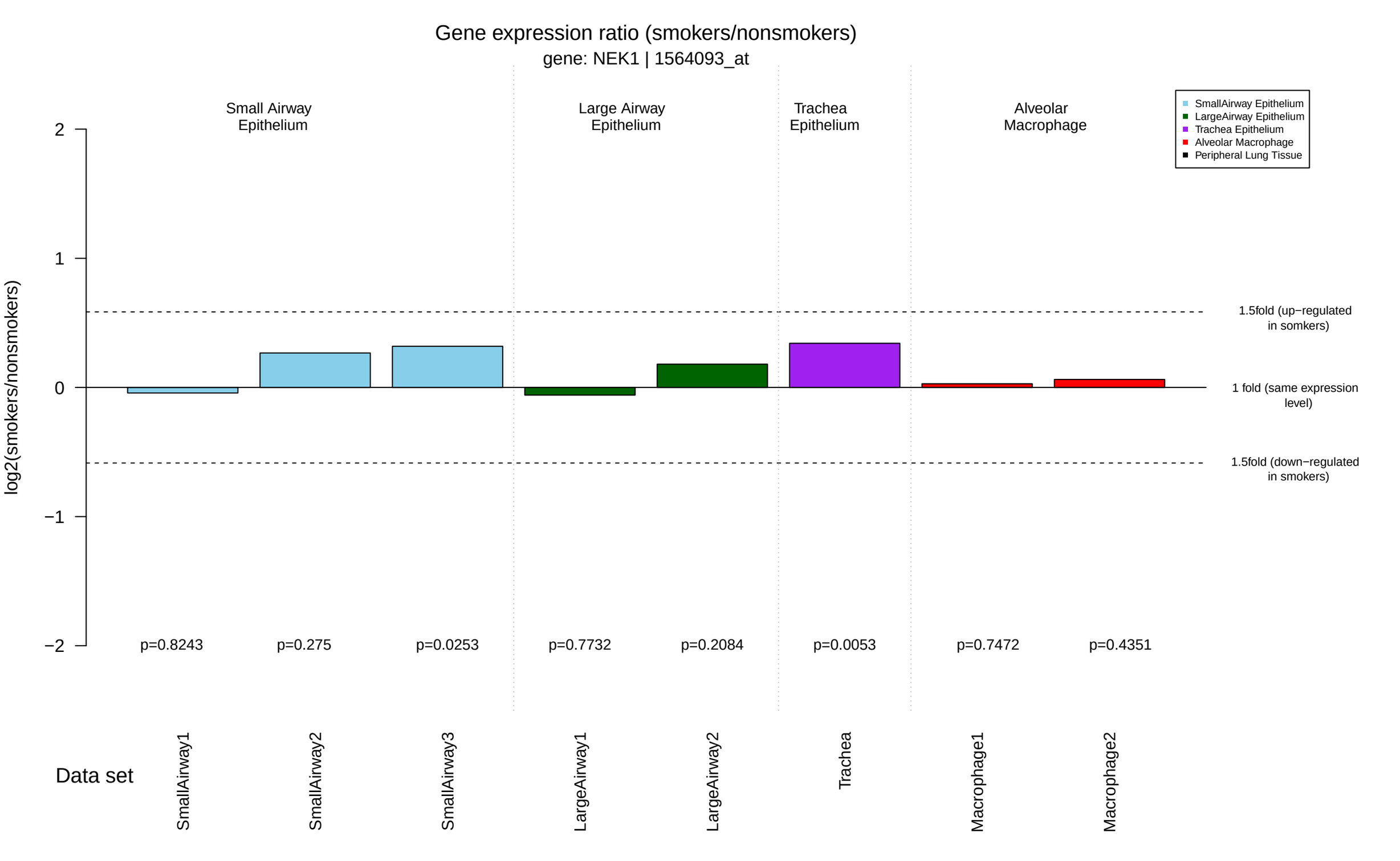
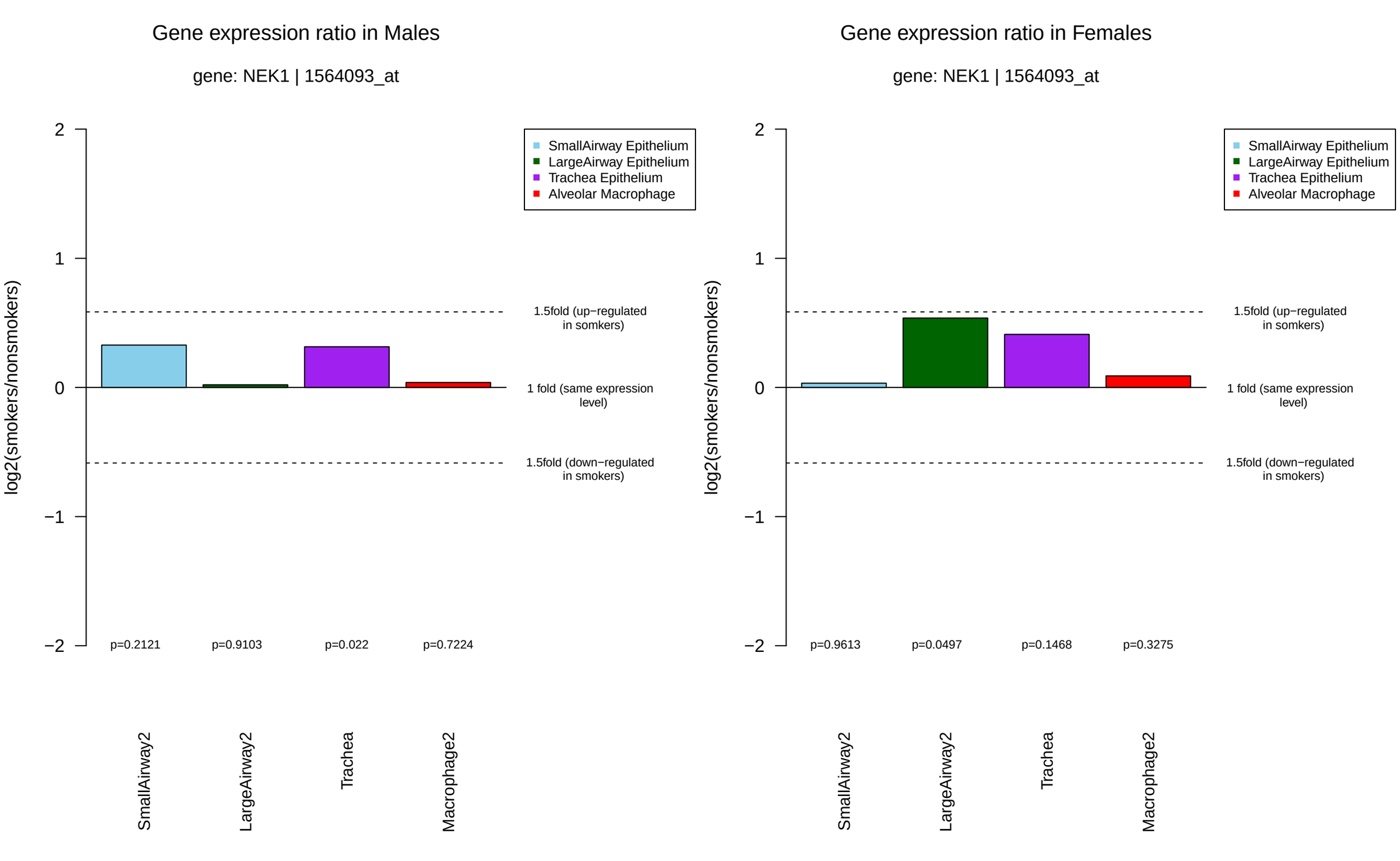
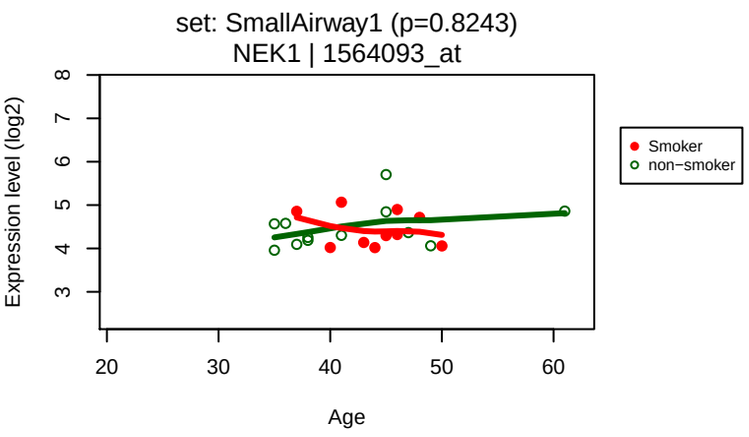
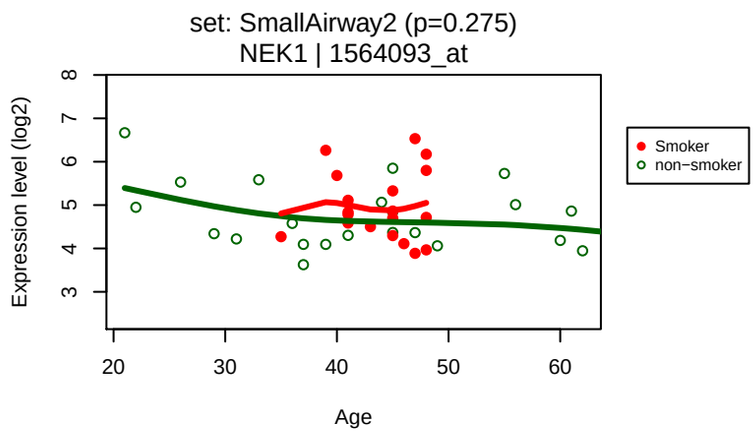
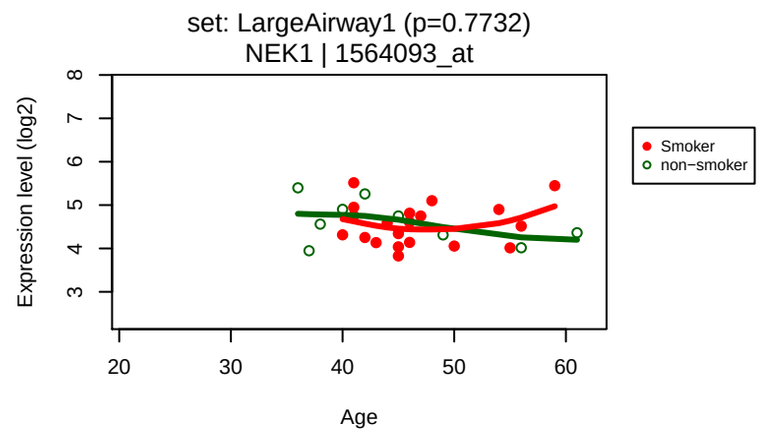

Gene expression of smokers and nonsmokers
gene: NEK1 | 1564093_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8243
p=0.275
p=0.0253
p=0.7732
p=0.2084
p=0.0053
p=0.7472
p=0.4351
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

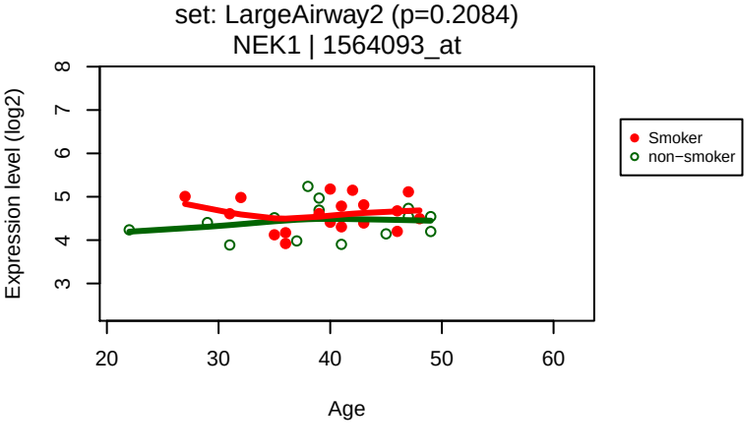

Gene expression ratio (smokers/nonsmokers)
gene: NEK1 | 1564093_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8243
p=0.275
p=0.0253
p=0.7732
p=0.2084
p=0.0053
p=0.7472
p=0.4351
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

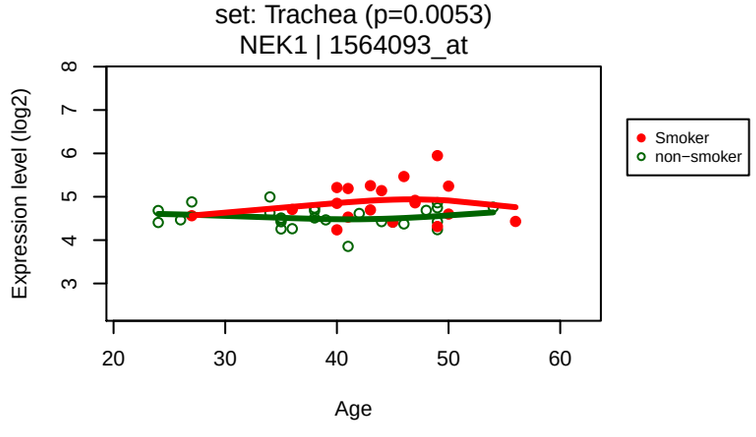

Gene expression ratio in Males
gene: NEK1 | 1564093_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2121
p=0.9103
p=0.022
p=0.7224
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK1 | 1564093_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9613
p=0.0497
p=0.1468
p=0.3275
SmallAirway2
LargeAirway2
Trachea
Macrophage2

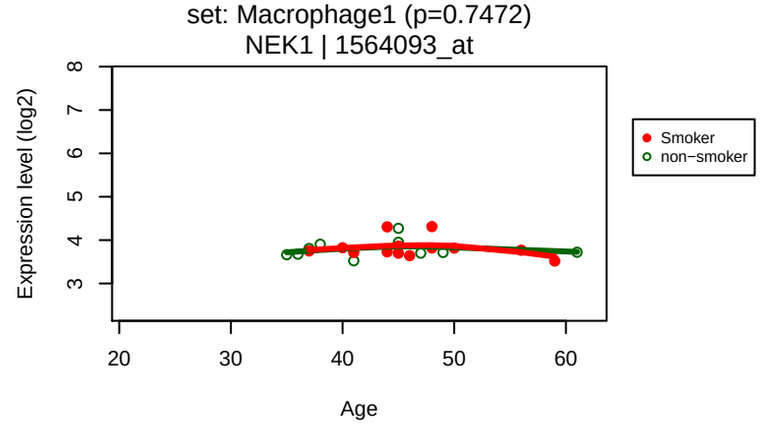

set: SmallAirway1 (p=0.8243)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker

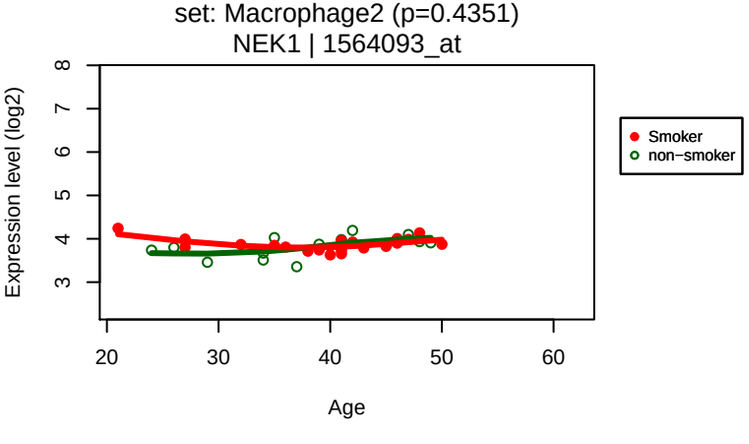

set: SmallAirway2 (p=0.275)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker

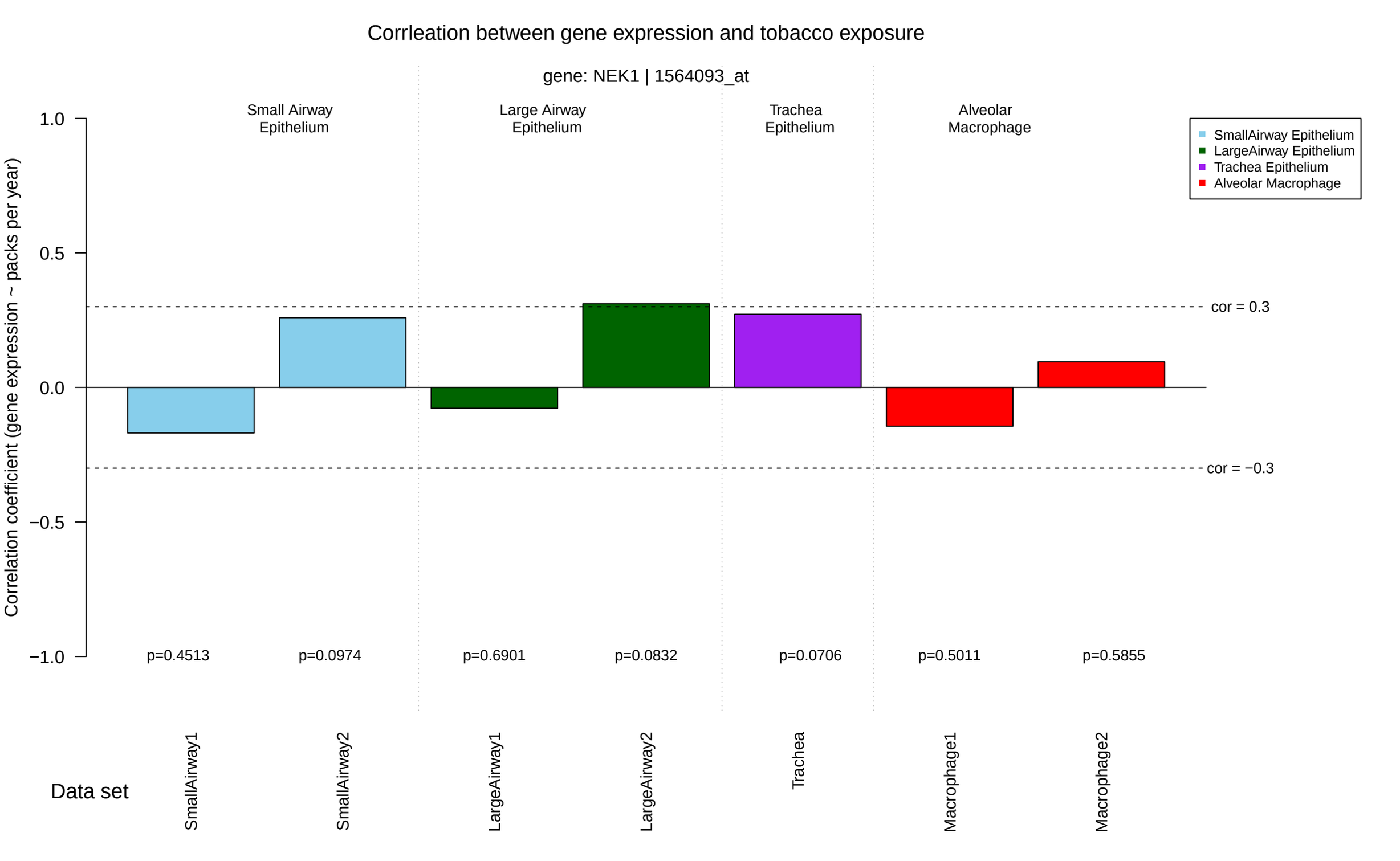

set: LargeAirway1 (p=0.7732)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2084)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0053)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7472)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4351)
NEK1 | 1564093_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK1 | 1564093_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4513
p=0.0974
p=0.6901
p=0.0832
p=0.0706
p=0.5011
p=0.5855
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2